

① Basic Execution Time Model $\Rightarrow$ ①
 It is based on execution time. It is assumed that failure may occur according to a non-homogenous poisson process (NHPP)
 Example of poisson process

- ① no. of telephone calls expected in a given period of time.
- ② expected number of roads accidents in given period of time

In this case, processes are non-homogenous, since failure intensity changes as a function of time.

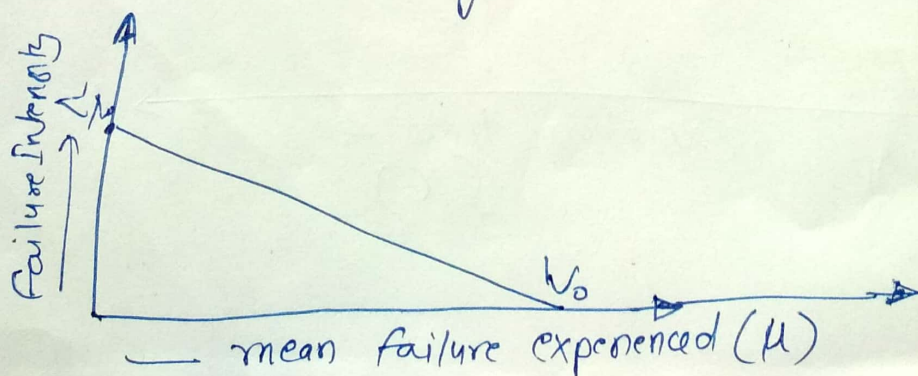
In this model, the decrease in failure intensity, as a function of the number of failures observed, is constant and is given by

$$\lambda(u) = \lambda_0 \left(1 - \frac{u}{V_0} \right)$$

λ_0 — Initial failure intensity at the start of execution

V_0 — no. of failure experienced, if program is executed for infinite time period.

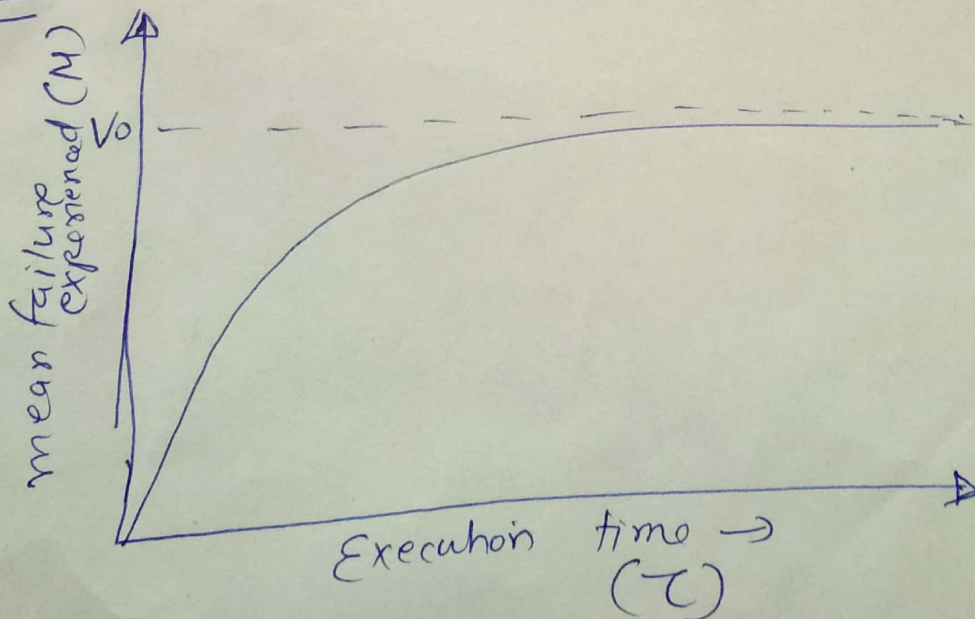
u — Avg or expected no. of failure exp at a given point in time.



The slope of the failure Intensity can be (2)
obtained by finding its first derivative

$$\frac{d\lambda}{dM} = \frac{-\lambda_0}{V_0}$$

This model implies a uniform operational
profile

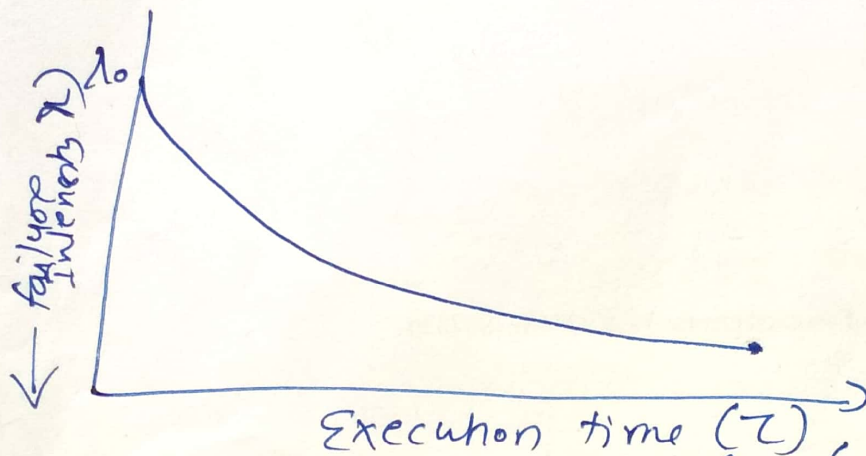


$$\frac{dM(z)}{dz} = \lambda_0 \left[1 - \frac{M(z)}{V_0} \right]$$

and above equation can be solved for $M(z)$

$$M(z) = V_0 \left[1 - \exp\left(\frac{-\lambda_0 z}{V_0}\right) \right]$$

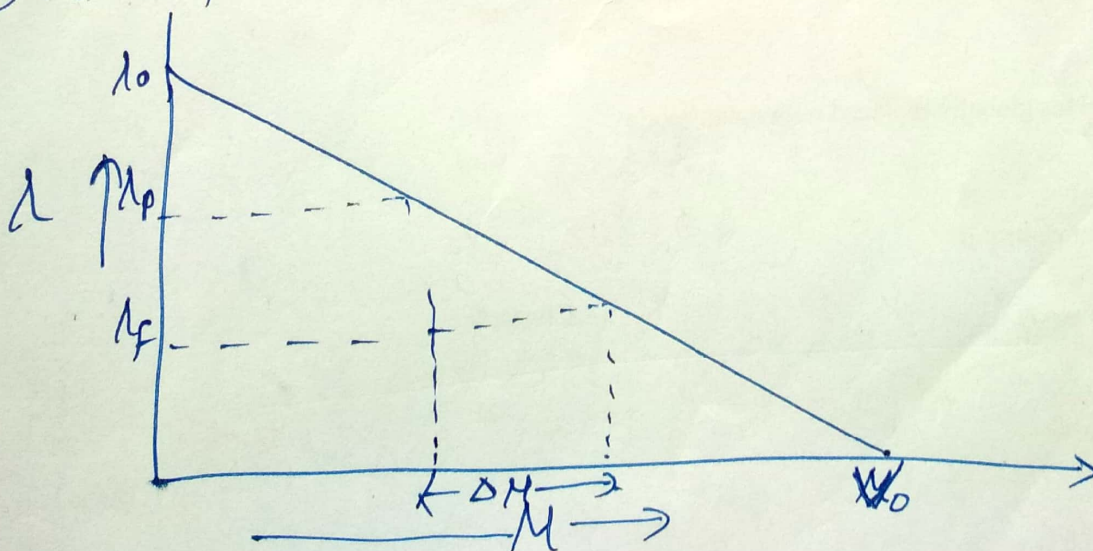
The failure Intensity as a function of execution time.



The relationship is useful for determining the present failure Intensity at any given value of execution time.

$$\lambda(z) = \lambda_0 \exp\left(\frac{-\lambda_0 z}{V_0}\right)$$

Derived Quantities: $\rightarrow$

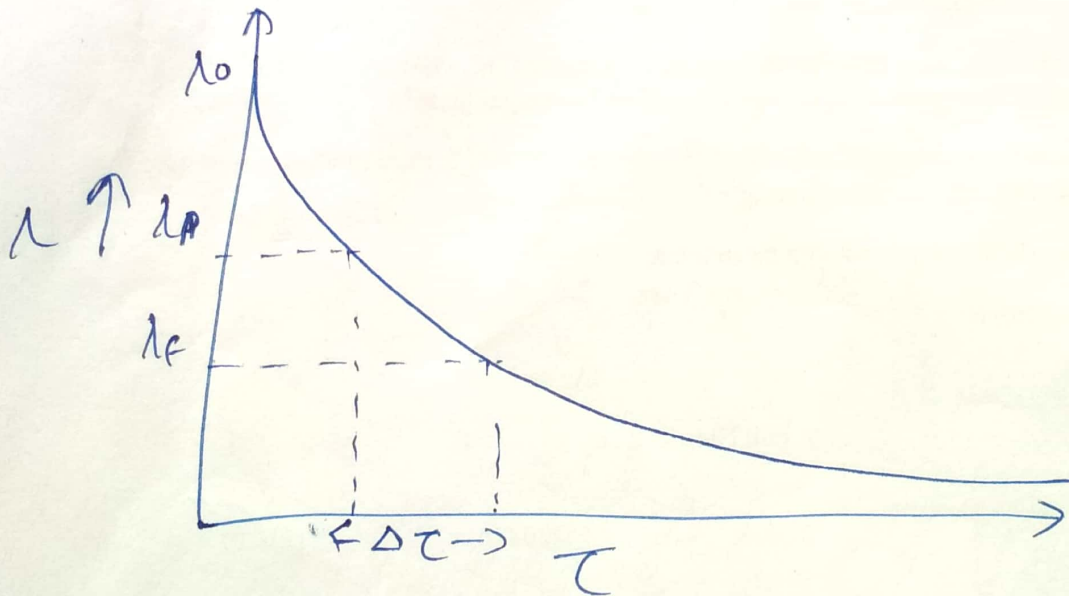


(3)

λ_0 → Initial Failure Intensity
 λ_p → Present " "
 λ_f → failure Intensity objective
 ΔM — Expected no. of additional failures to be experienced to reach failure Intensity objective

$$\Delta M = \frac{V_0}{\lambda_0} (\lambda_p - \lambda_f)$$

Similarly we can determine the additional execution time ΔT required to reach the failure Intensity objective



$$\Delta T = \frac{V_0}{\lambda_0} \ln \left[\frac{\lambda_p}{\lambda_f} \right]$$

Hence ΔT and ΔM give us idea about failure correction workload. Both are used in making estimates of the additional calendar time required for failure Intensity obj to reach the

Logarithmic Poisson Execution Time Model

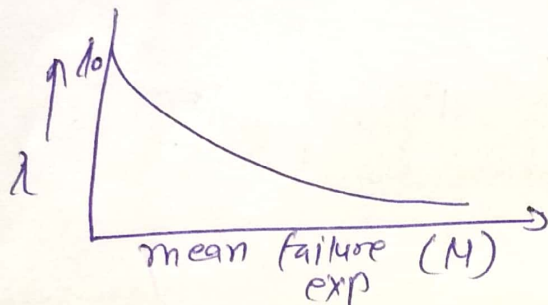
This model is developed by Musa et. al [793]

In this case, the failure intensity function decreases exponentially whereas it is constant for basic model.

The failure Intensity function

$$\lambda(M) = \lambda_0 \exp(-\theta M)$$

Where θ is failure intensity decay parameter



The ' θ ' represents the relative change of failure intensity per failure experienced. The slope of failure intensity function is

$$\frac{d\lambda}{dM} = -\lambda_0 \theta \exp(-M\theta)$$

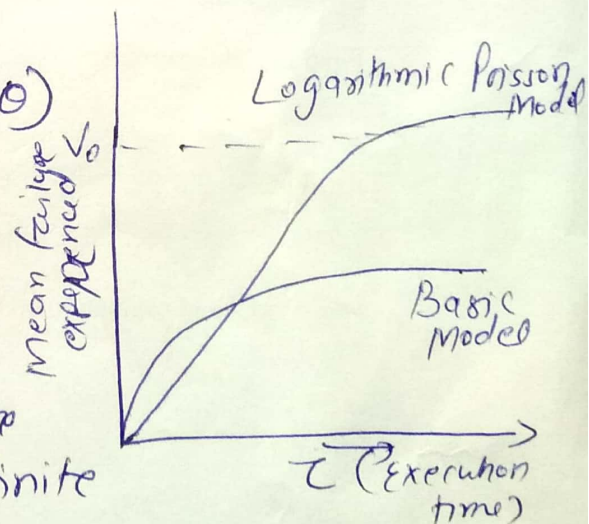
$$\frac{d\lambda}{dM} = -\theta \lambda$$

The expected number of failure for this model is always infinite at infinite time.

$$M(\tau) = \frac{1}{\theta} \ln(\lambda_0 \theta \tau + 1)$$

$$\lambda(\tau) = \lambda_0 (\lambda_0 \theta \tau + 1)$$

The relationship for the additional no. of failures and additional execution time in this model



$$\Delta M = \frac{1}{\theta} \ln \left(\frac{\lambda_P}{\lambda_F} \right)$$

(11)

$$\Delta Z = \frac{1}{\theta} \left[\frac{1}{\lambda_F} - \frac{1}{\lambda_P} \right]$$

Hence at larger values of execution time, the logarithmic poisson model will have larger values of failure intensity than Basic model.

Calendar Time Component : $\Rightarrow$

The calendar time component is based on a debugging process model.

- ① Resource used in operating the program for a given execution time and processing an associated quantity of failures
- ② Resource quantities available
- ③ the degree to which a resource can be utilized (due to bottleneck) during the period in which it is limiting.

Resource Usage: $\Rightarrow$

(6)

~~Res~~ let X_r be the usage of resource r , then

$$X_r = \theta_r Z + M_r M$$

θ_r — Resource Usage per CPU hr. It is nonzero for failure identification personnel (θ_I) and computer time (θ_c).

M_r — Resource Usage per failure.

Resource	Usage Parameters requirement per		Planned Parameters	
	CPU hour	Failure	Quantities available	Utilization
Failure Identification	θ_I	MI	PI	I
Failure correction	O	MF	PF	Pf
Computer time	θ_c	Mc	Pc	Pc

$$X_c = M_c \Delta U + \theta_c \Delta Z \quad \text{for compute Time}$$

$$X_f = M_f \Delta M \quad \text{for failure correction}$$

$$X_I = M_I \Delta M + \theta_I \Delta Z \quad \text{for failure Identification}$$